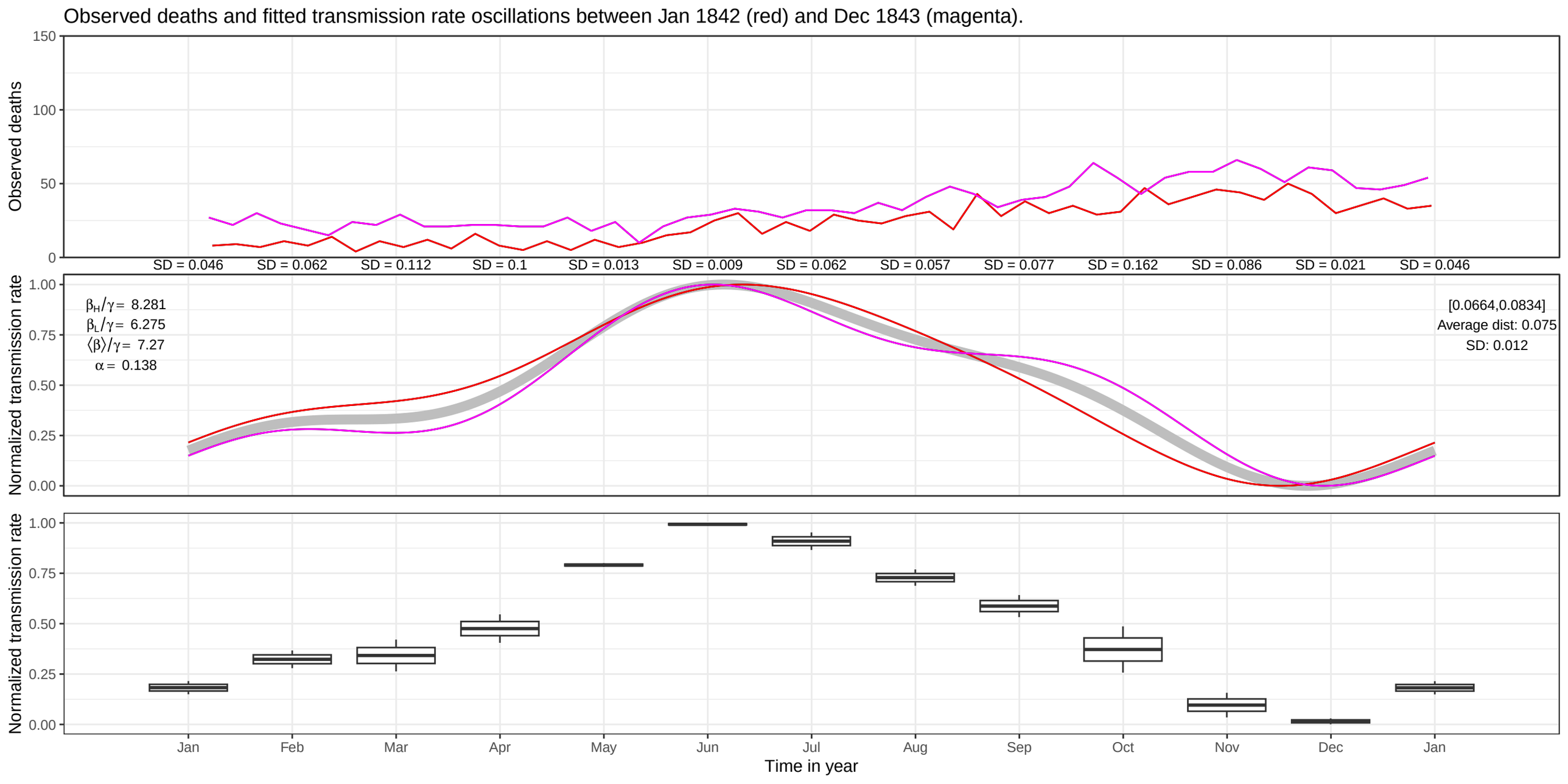
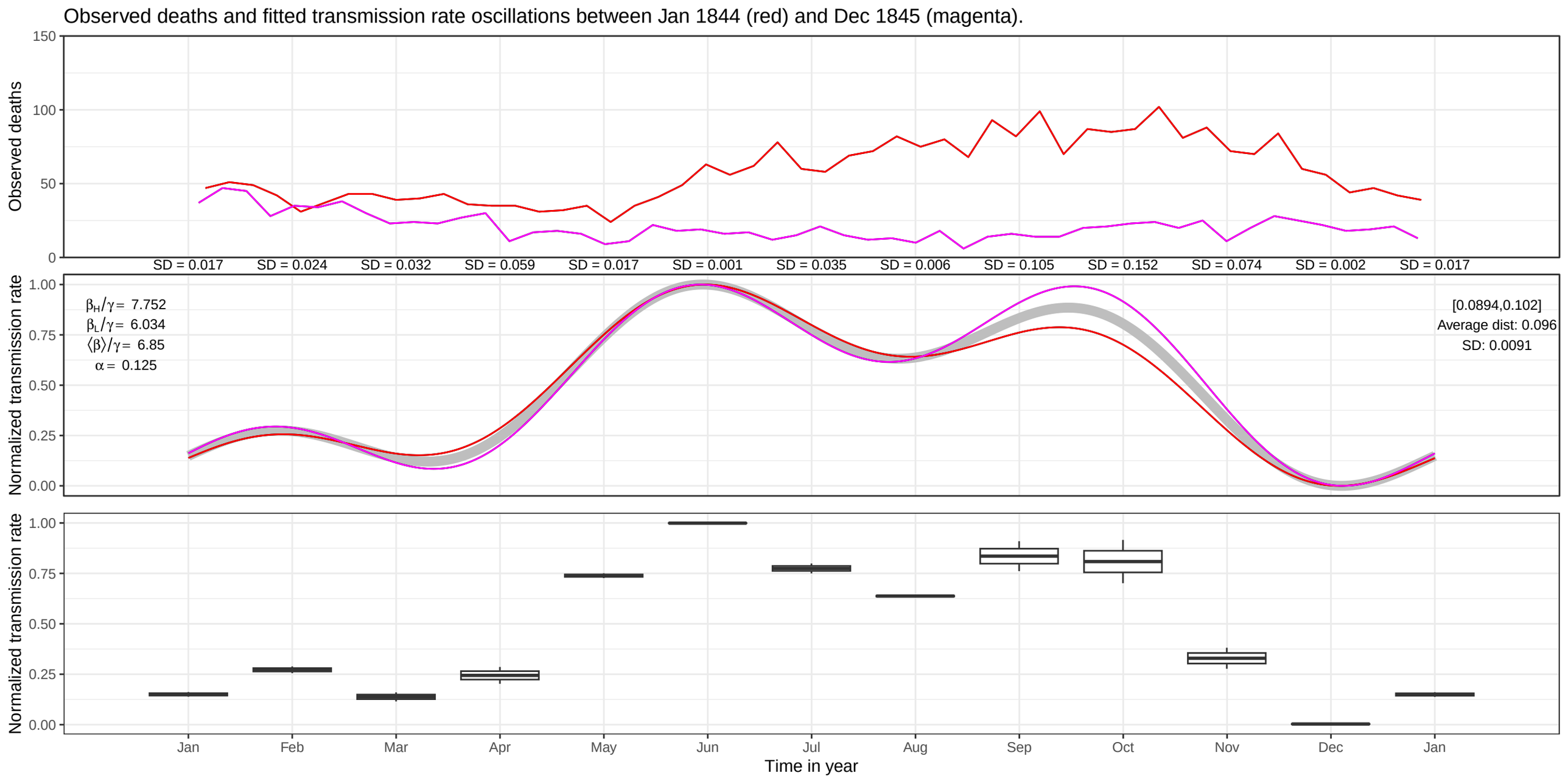
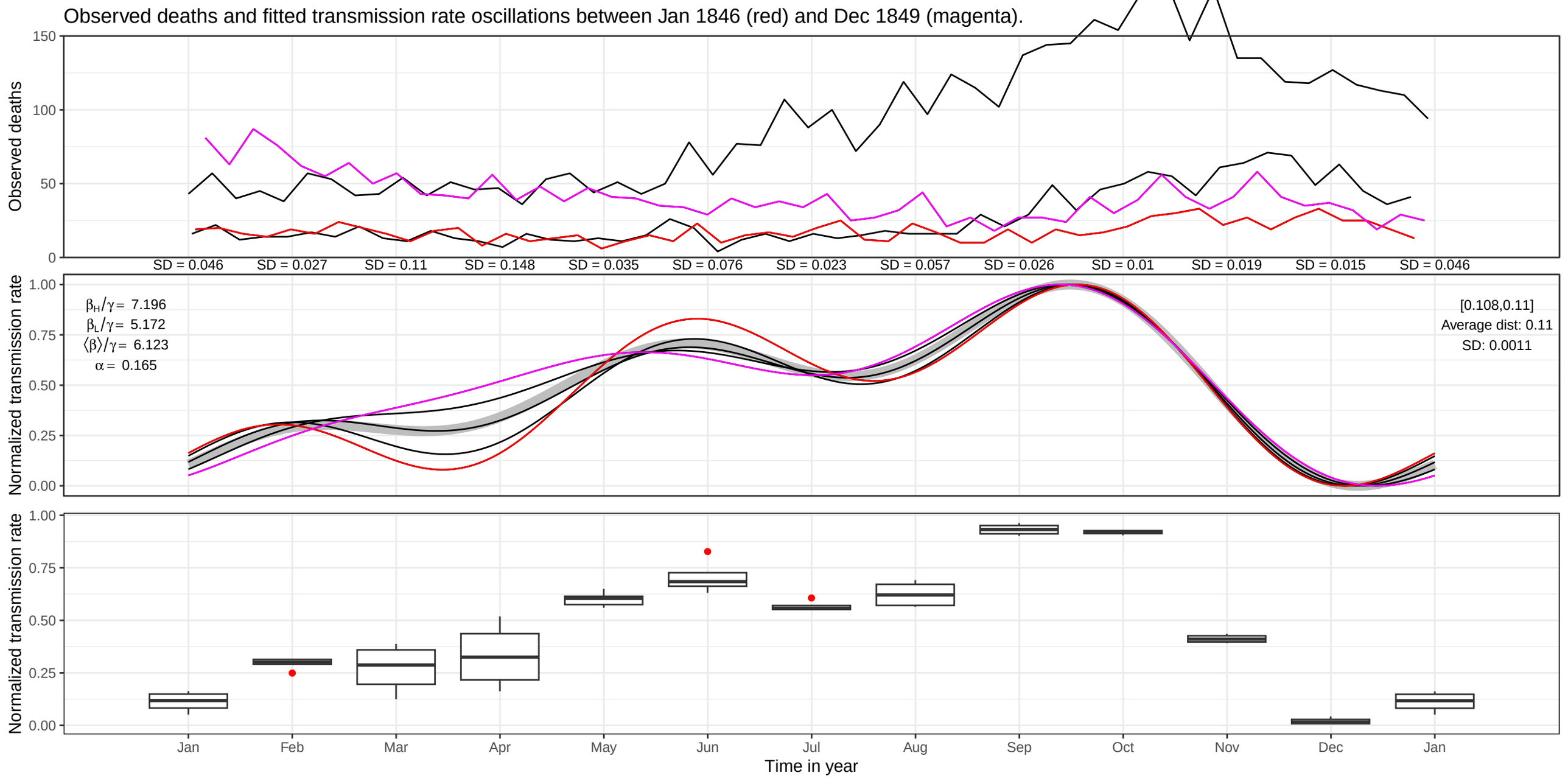
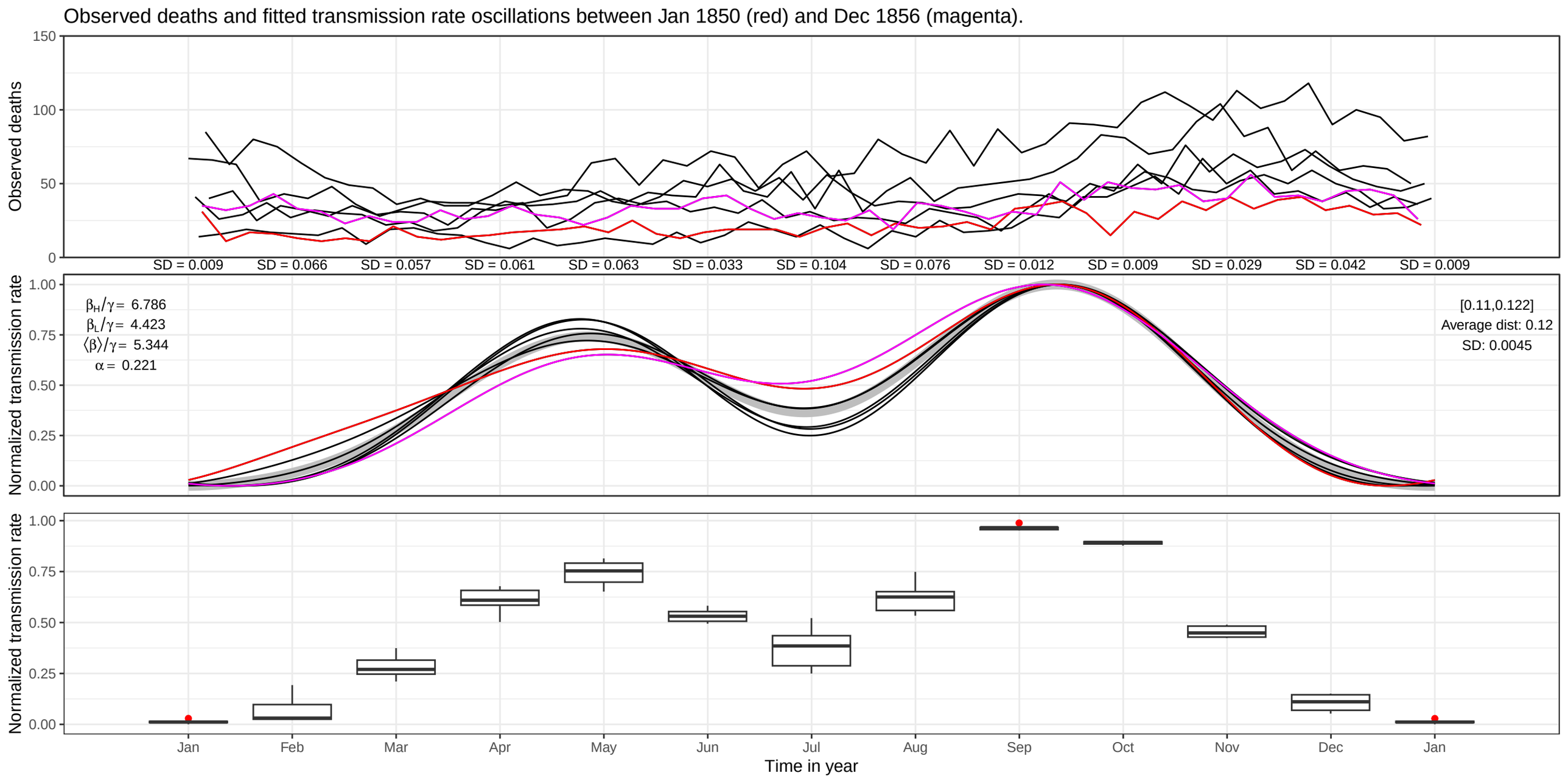
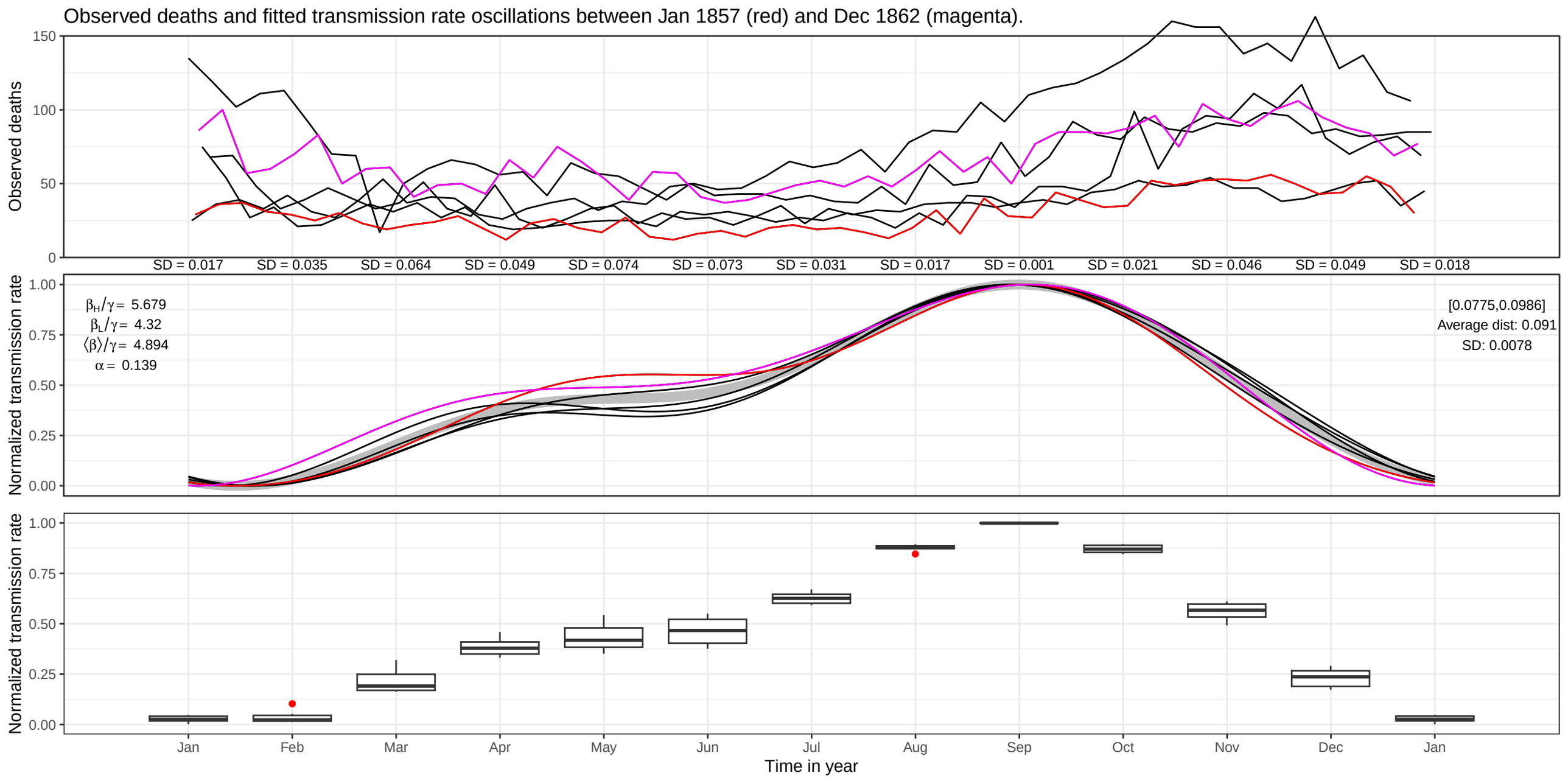
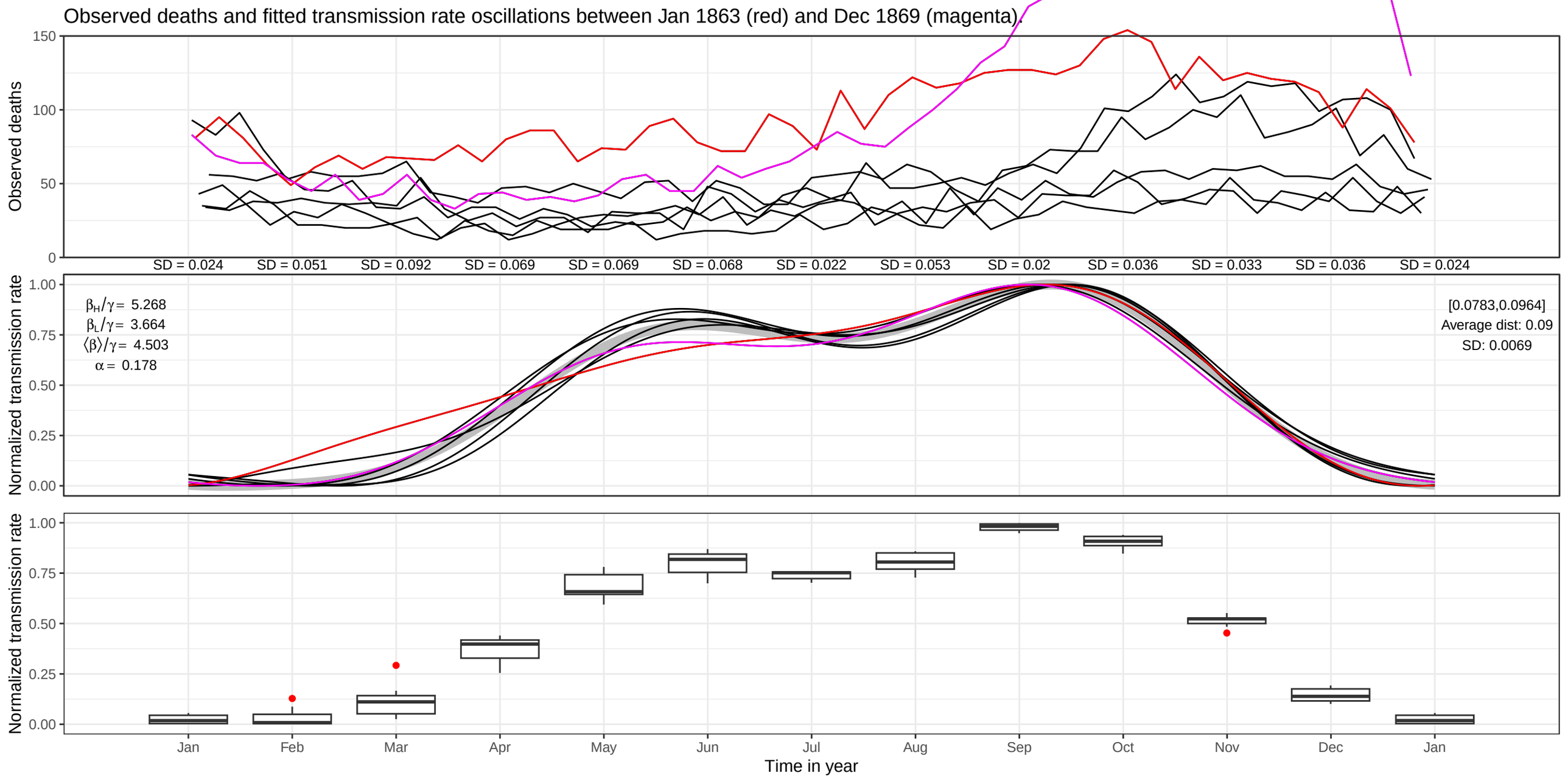

Observed deaths and fitted transmission rate oscillations between Jan 1842 (red) and Dec 1843 (magenta).
Observed deaths
150
100
50
0
SD = 0.046
SD = 0.062
SD = 0.112
SD = 0.1
SD = 0.013
SD = 0.009
SD = 0.062
SD = 0.057
SD = 0.077
SD = 0.162
SD = 0.086
SD = 0.021
SD = 0.046
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 8.281
βL/γ = 6.275
⟨β⟩/γ = 7.27
α = 0.138
[0.0664,0.0834]
Average dist: 0.075
SD: 0.012
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

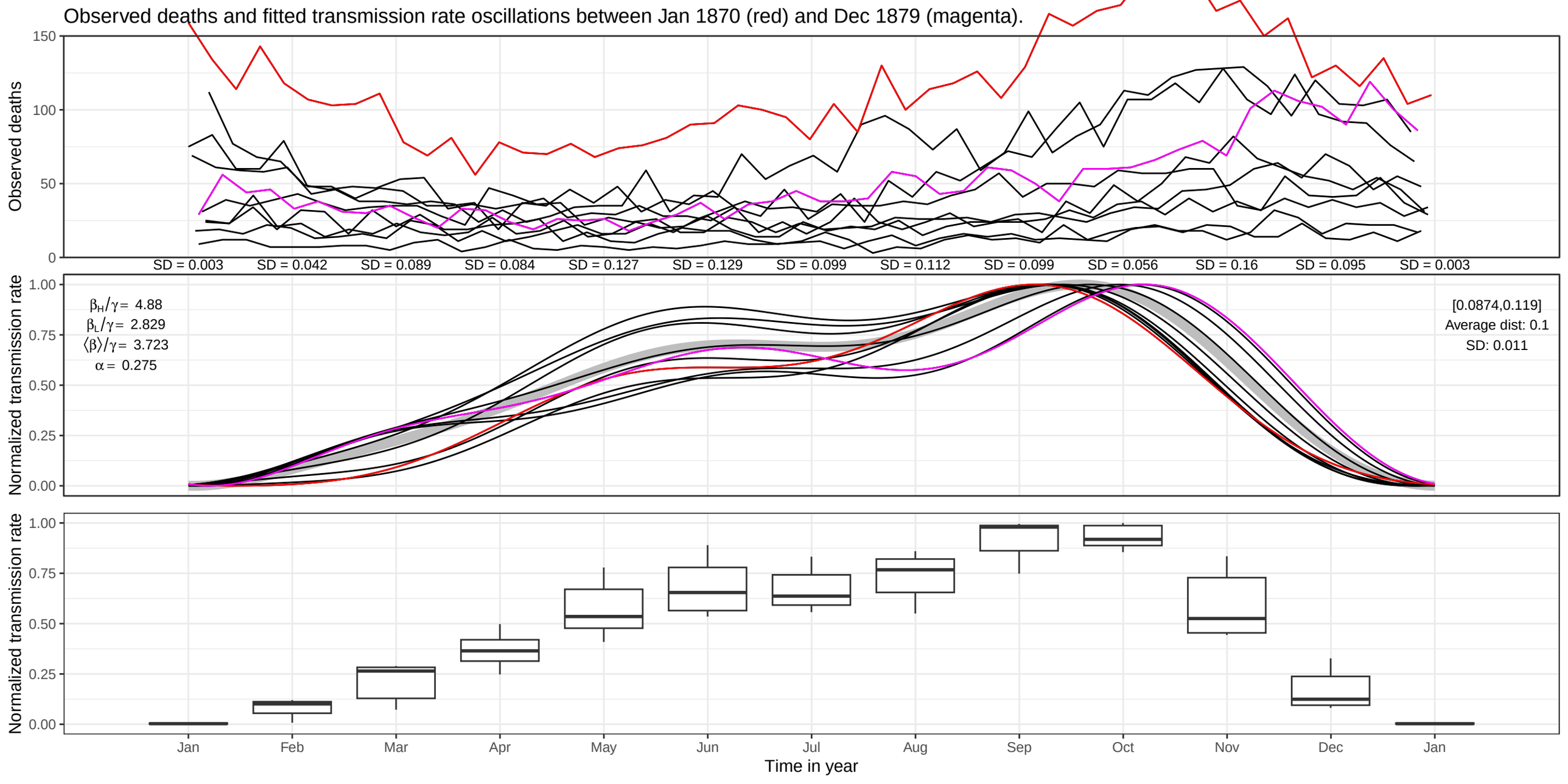

Observed deaths and fitted transmission rate oscillations between Jan 1844 (red) and Dec 1845 (magenta).
Observed deaths
150
100
50
0
SD = 0.017
SD = 0.024
SD = 0.032
SD = 0.059
SD = 0.017
SD = 0.001
SD = 0.035
SD = 0.006
SD = 0.105
SD = 0.152
SD = 0.074
SD = 0.002
SD = 0.017
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 7.752
βL/γ = 6.034
⟨β⟩/γ = 6.85
α = 0.125
[0.0894,0.102]
Average dist: 0.096
SD: 0.0091
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

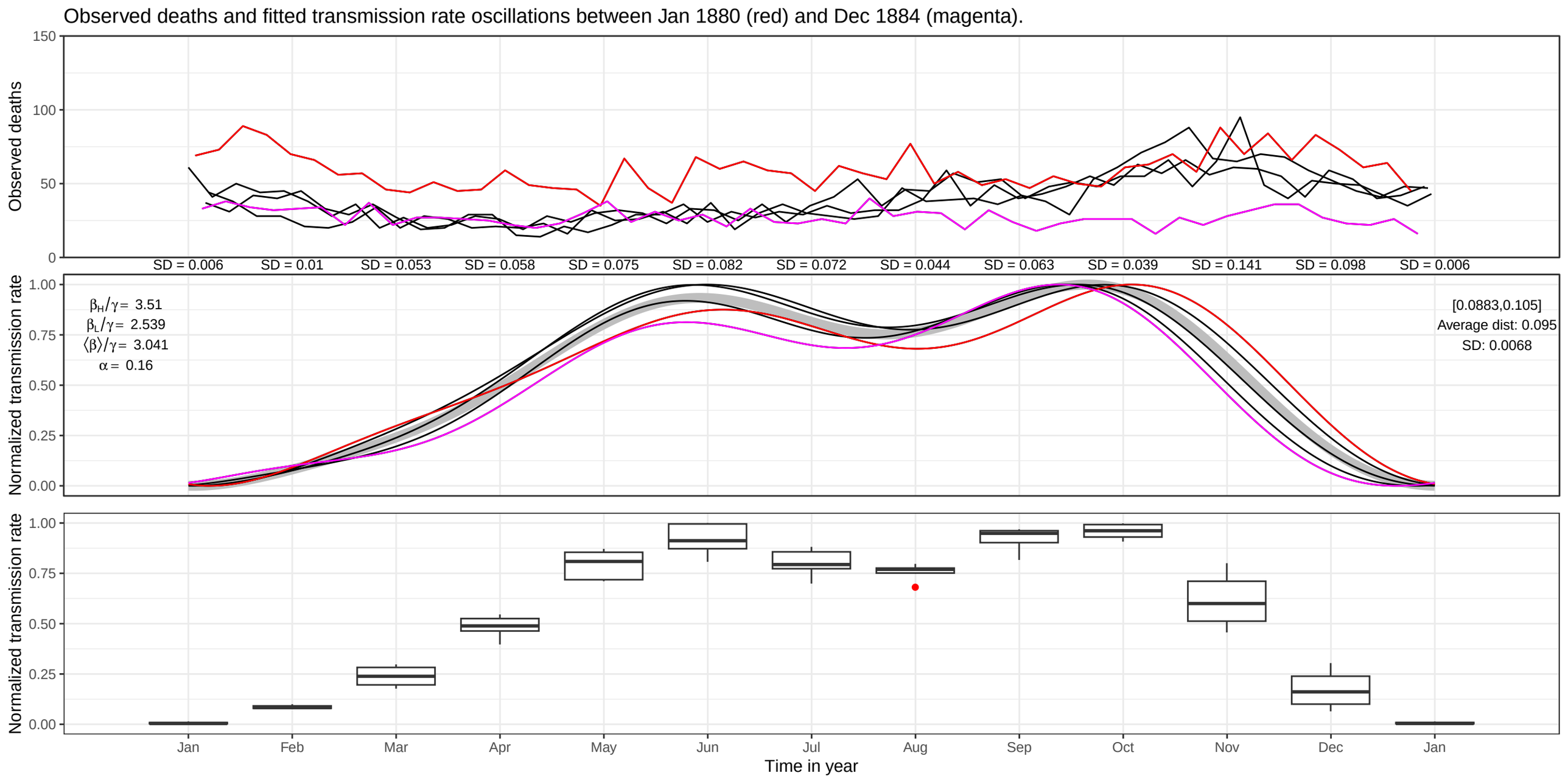

Observed deaths and fitted transmission rate oscillations between Jan 1846 (red) and Dec 1849 (magenta).
Observed deaths
150
100
50
0
SD = 0.046
SD = 0.027
SD = 0.11
SD = 0.148
SD = 0.035
SD = 0.076
SD = 0.023
SD = 0.057
SD = 0.026
SD = 0.01
SD = 0.019
SD = 0.015
SD = 0.046
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 7.196
βL/γ = 5.172
⟨β⟩/γ = 6.123
α = 0.165
[0.108,0.11]
Average dist: 0.11
SD: 0.0011
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

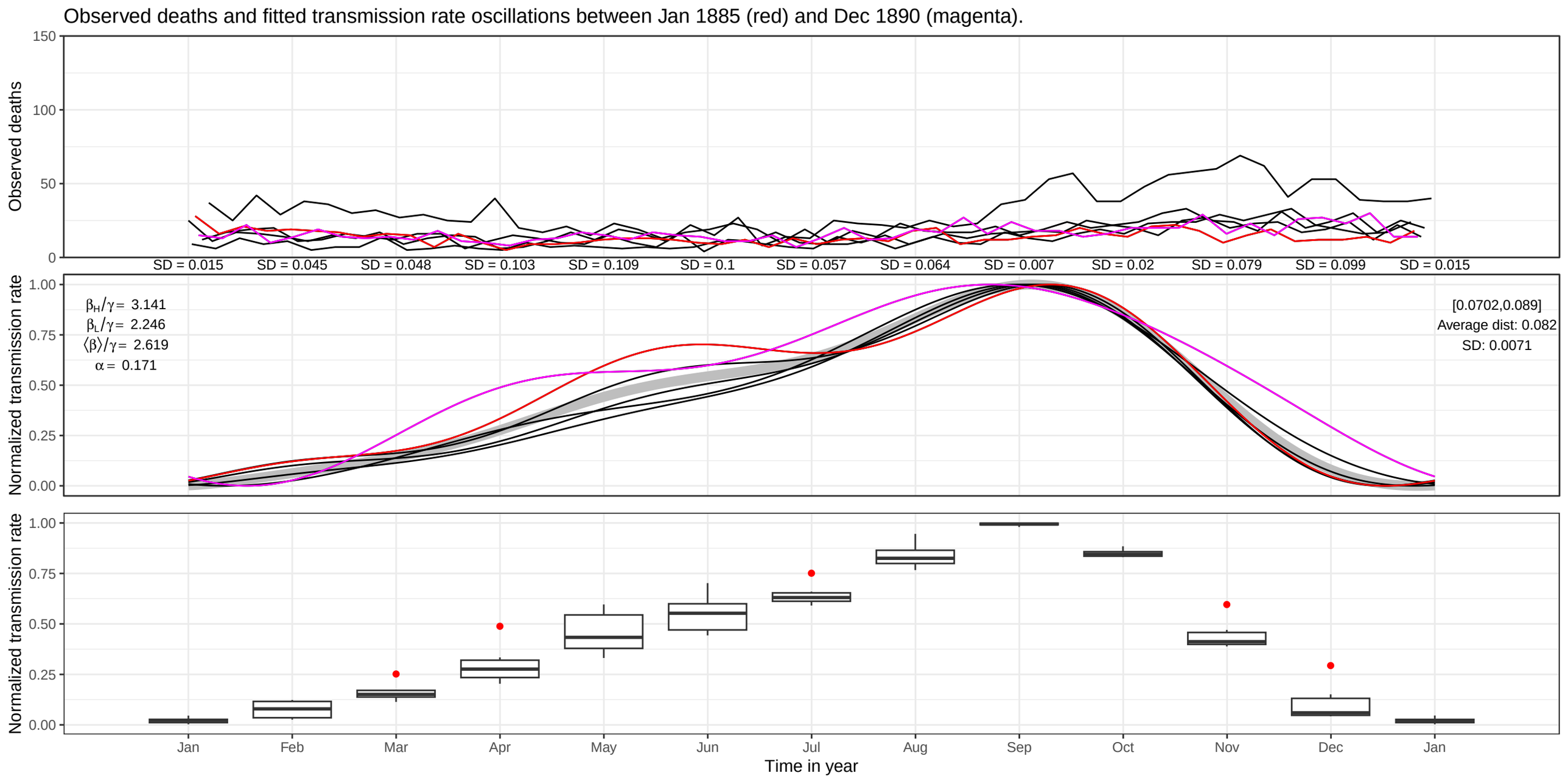

Observed deaths and fitted transmission rate oscillations between Jan 1850 (red) and Dec 1856 (magenta).
Observed deaths
150
100
50
0
SD = 0.009
SD = 0.066
SD = 0.057
SD = 0.061
SD = 0.063
SD = 0.033
SD = 0.104
SD = 0.076
SD = 0.012
SD = 0.009
SD = 0.029
SD = 0.042
SD = 0.009
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 6.786
βL/γ = 4.423
⟨β⟩/γ = 5.344
α = 0.221
[0.11,0.122]
Average dist: 0.12
SD: 0.0045
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

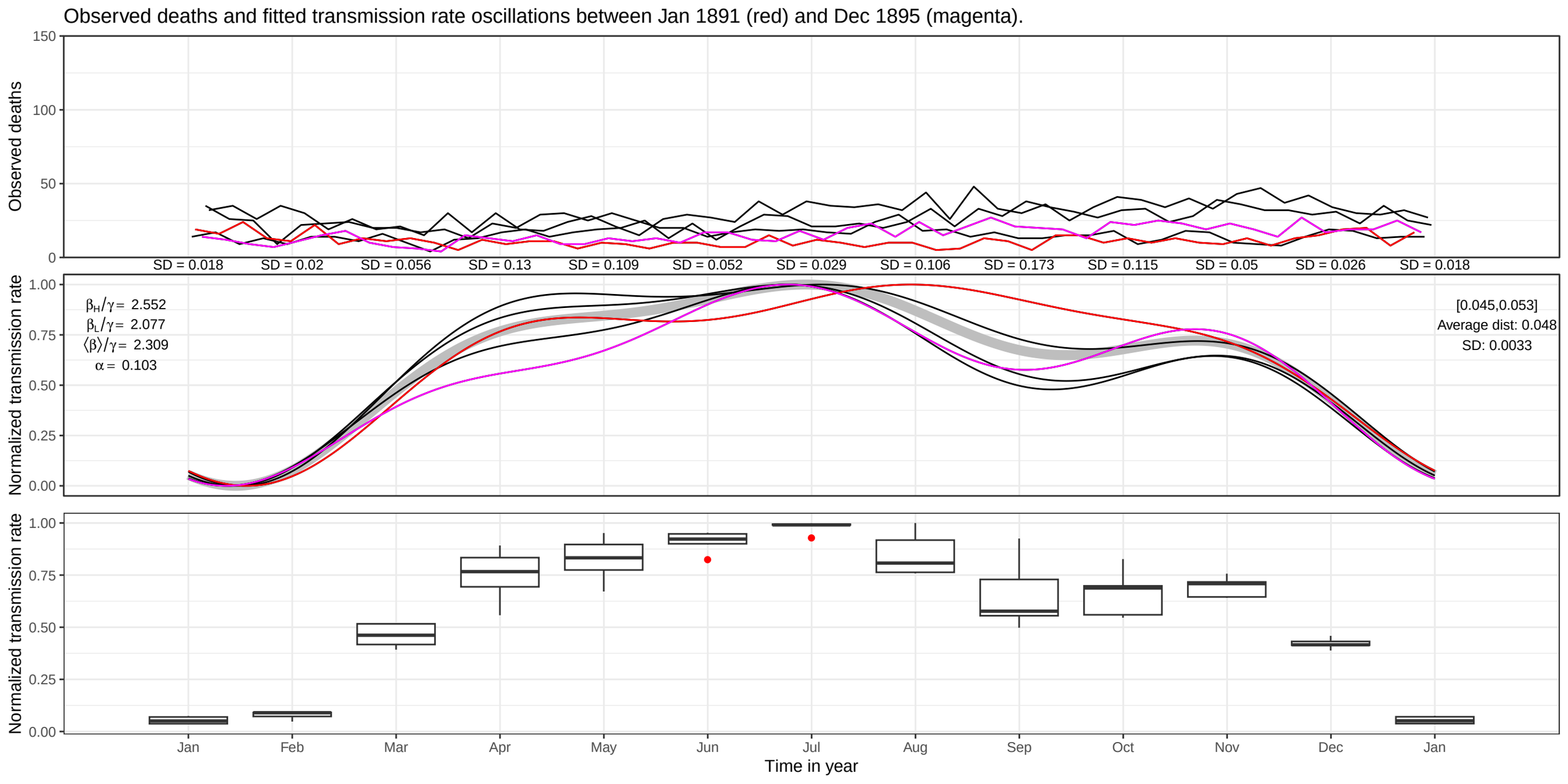

Observed deaths and fitted transmission rate oscillations between Jan 1857 (red) and Dec 1862 (magenta).
Observed deaths
150
100
50
0
SD = 0.017
SD = 0.035
SD = 0.064
SD = 0.049
SD = 0.074
SD = 0.073
SD = 0.031
SD = 0.017
SD = 0.001
SD = 0.021
SD = 0.046
SD = 0.049
SD = 0.018
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 5.679
βL/γ = 4.32
⟨β⟩/γ = 4.894
α = 0.139
[0.0775,0.0986]
Average dist: 0.091
SD: 0.0078
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

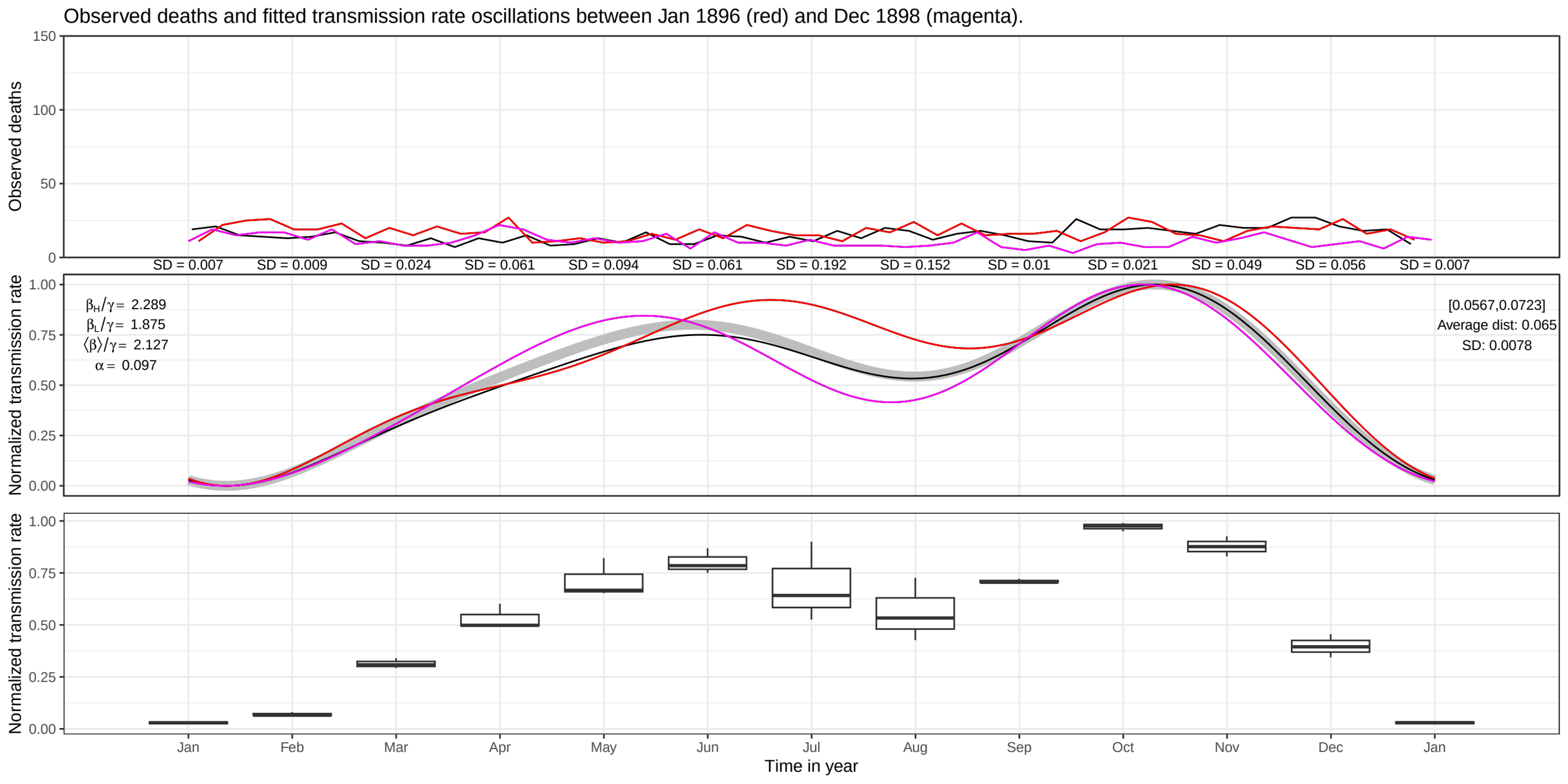

Observed deaths and fitted transmission rate oscillations between Jan 1863 (red) and Dec 1869 (magenta).
Observed deaths
150
100
50
0
SD = 0.024
SD = 0.051
SD = 0.092
SD = 0.069
SD = 0.069
SD = 0.068
SD = 0.022
SD = 0.053
SD = 0.02
SD = 0.036
SD = 0.033
SD = 0.036
SD = 0.024
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 5.268
βL/γ = 3.664
⟨β⟩/γ = 4.503
α = 0.178
[0.0783,0.0964]
Average dist: 0.09
SD: 0.0069
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

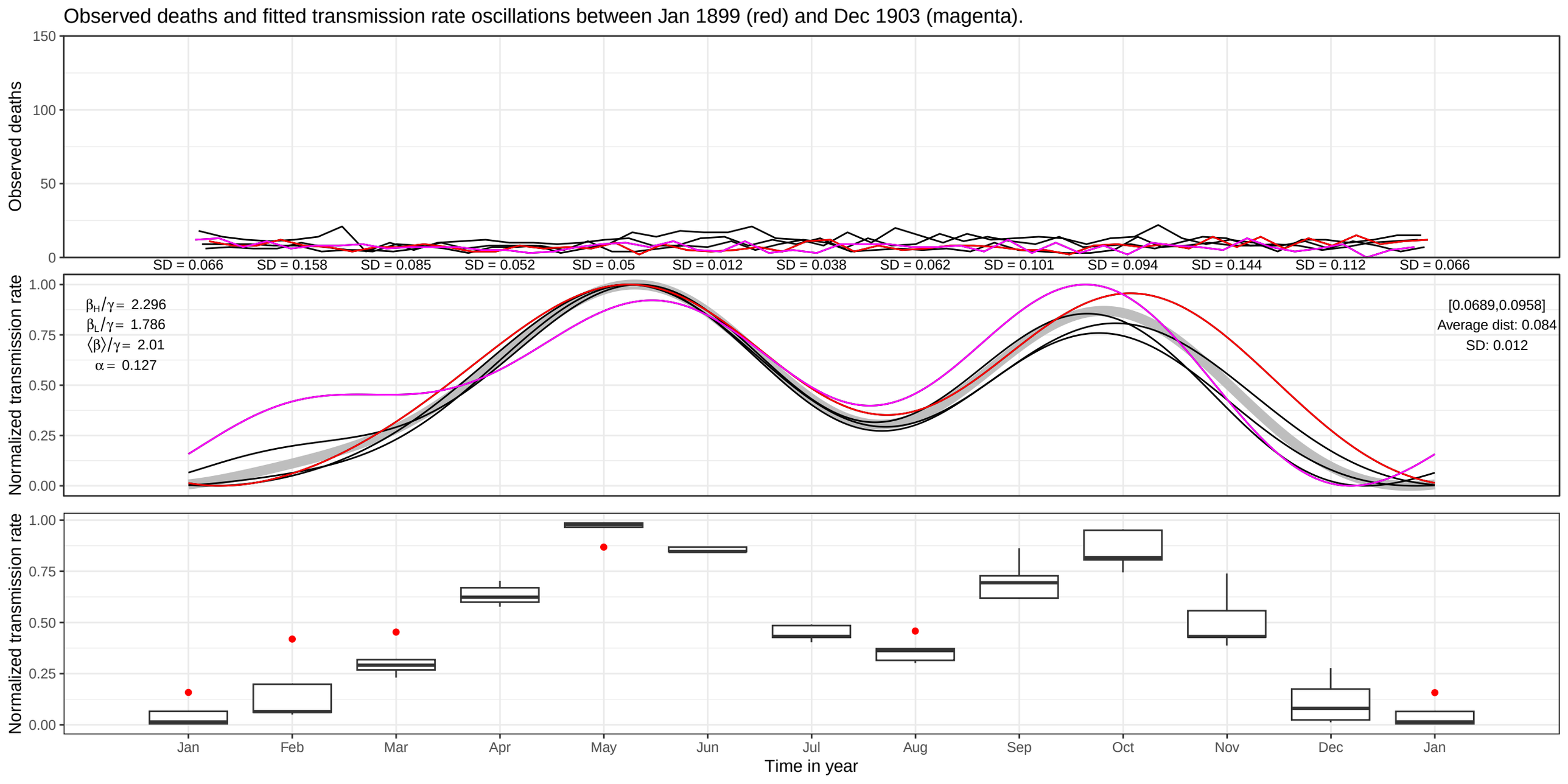

Observed deaths and fitted transmission rate oscillations between Jan 1870 (red) and Dec 1879 (magenta).
Observed deaths
150
100
50
0
SD = 0.003
SD = 0.042
SD = 0.089
SD = 0.084
SD = 0.127
SD = 0.129
SD = 0.099
SD = 0.112
SD = 0.099
SD = 0.056
SD = 0.16
SD = 0.095
SD = 0.003
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 4.88
$\beta_L/\gamma =$ 2.829
$\langle\beta\rangle/\gamma =$ 3.723
$\alpha =$ 0.275
[0.0874,0.119]
Average dist: 0.1
SD: 0.011
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

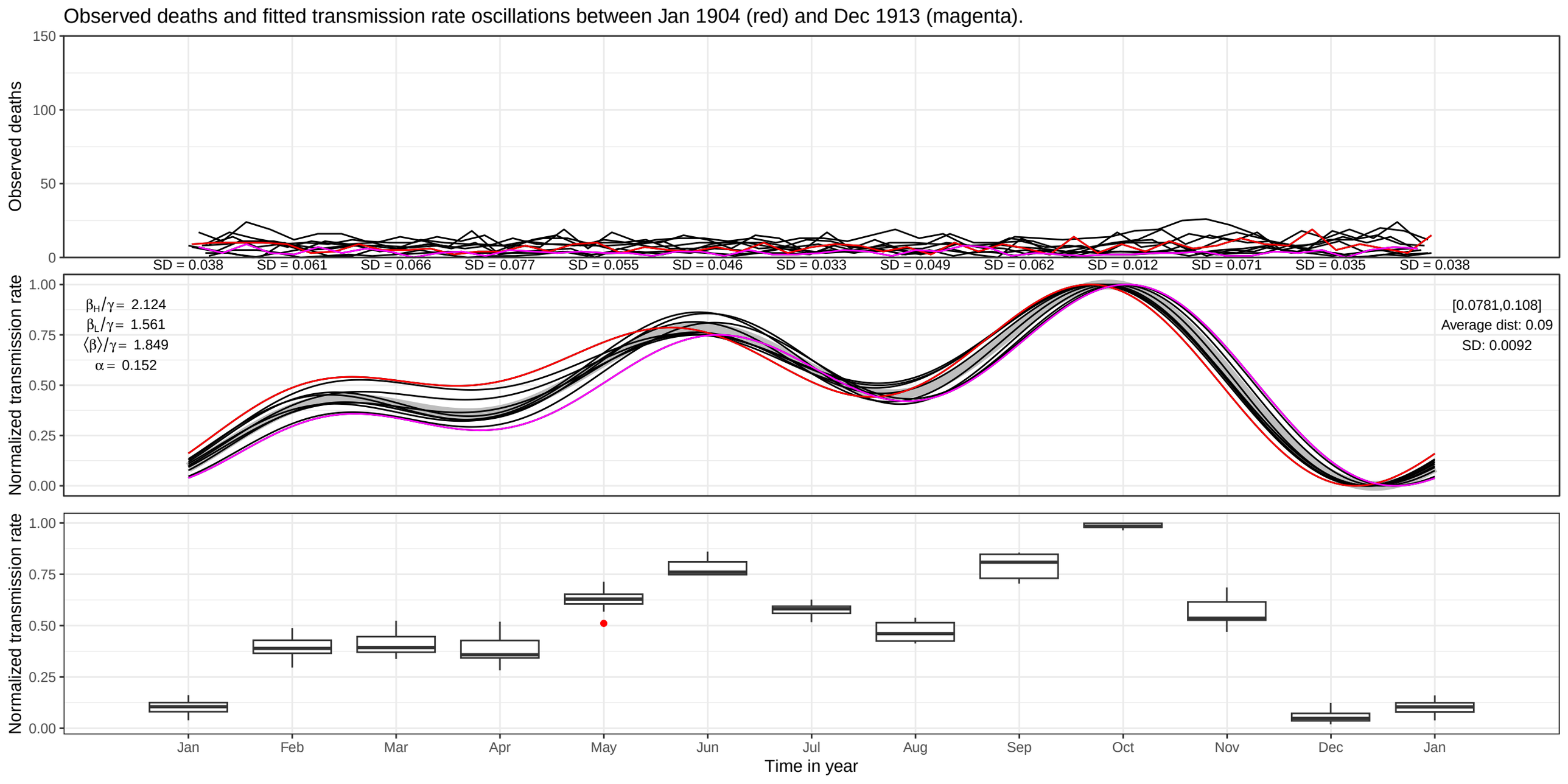

Observed deaths and fitted transmission rate oscillations between Jan 1880 (red) and Dec 1884 (magenta).
Observed deaths
150
100
50
0
SD = 0.006
SD = 0.01
SD = 0.053
SD = 0.058
SD = 0.075
SD = 0.082
SD = 0.072
SD = 0.044
SD = 0.063
SD = 0.039
SD = 0.141
SD = 0.098
SD = 0.006
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 3.51
βL/γ = 2.539
⟨β⟩/γ = 3.041
α = 0.16
[0.0883,0.105]
Average dist: 0.095
SD: 0.0068
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

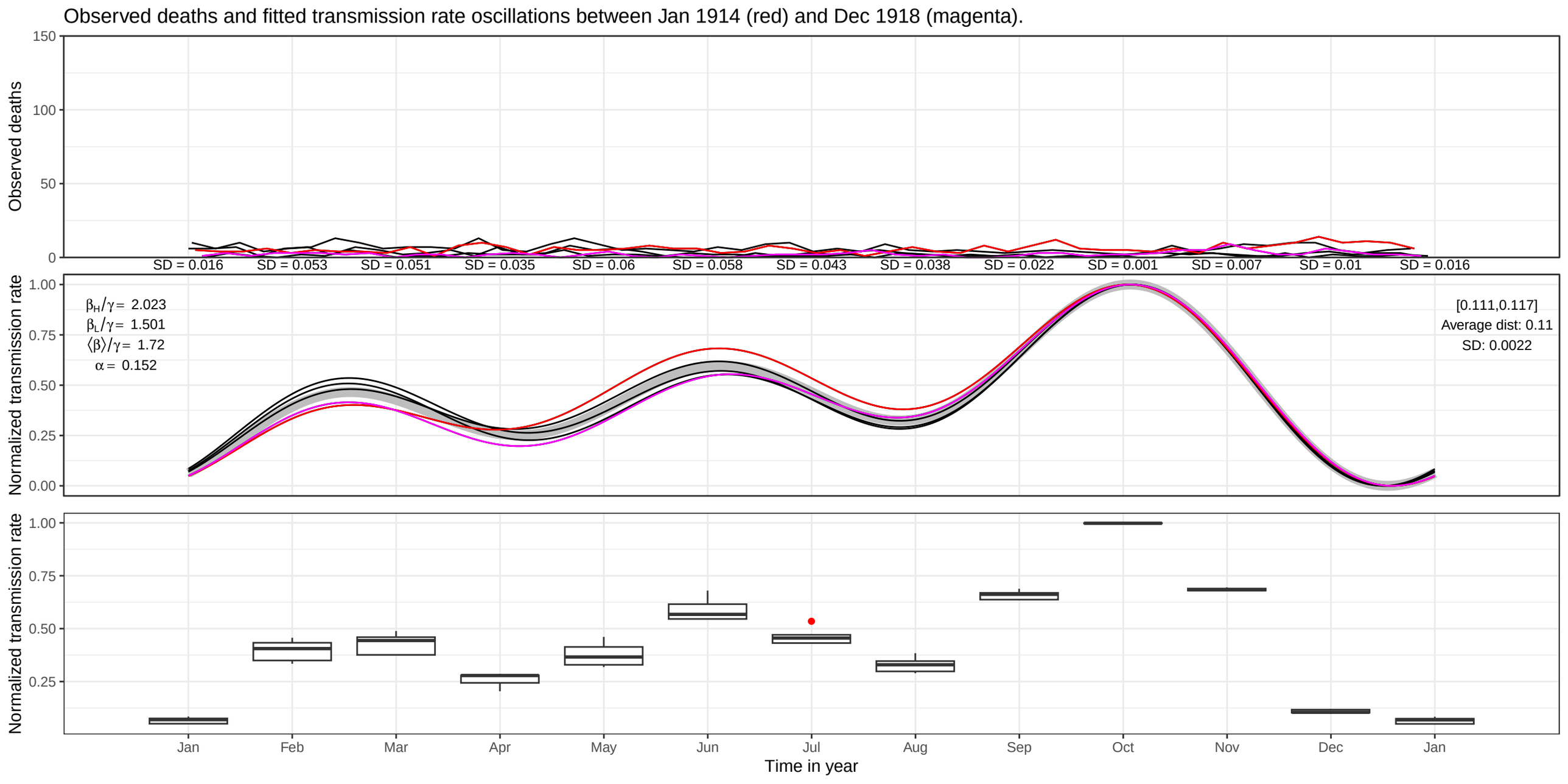

Observed deaths and fitted transmission rate oscillations between Jan 1885 (red) and Dec 1890 (magenta).
Observed deaths
150
100
50
0
SD = 0.015
SD = 0.045
SD = 0.048
SD = 0.103
SD = 0.109
SD = 0.1
SD = 0.057
SD = 0.064
SD = 0.007
SD = 0.02
SD = 0.079
SD = 0.099
SD = 0.015
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 3.141
βL/γ = 2.246
⟨β⟩/γ = 2.619
α = 0.171
[0.0702,0.089]
Average dist: 0.082
SD: 0.0071
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

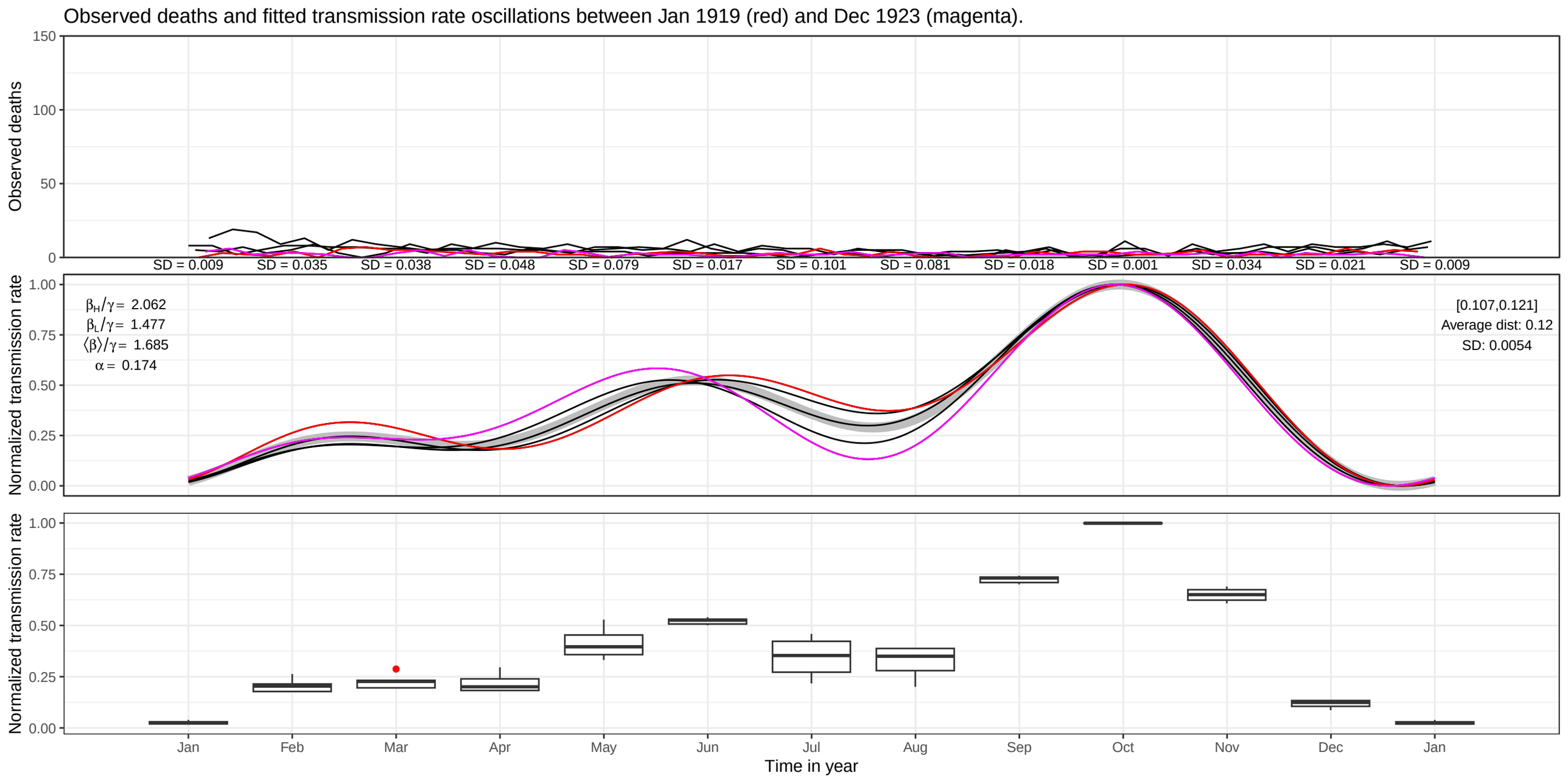

Observed deaths and fitted transmission rate oscillations between Jan 1891 (red) and Dec 1895 (magenta).
Observed deaths
150
100
50
0
SD = 0.018
SD = 0.02
SD = 0.056
SD = 0.13
SD = 0.109
SD = 0.052
SD = 0.029
SD = 0.106
SD = 0.173
SD = 0.115
SD = 0.05
SD = 0.026
SD = 0.018
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 2.552
βL/γ = 2.077
⟨β⟩/γ = 2.309
α = 0.103
[0.045,0.053]
Average dist: 0.048
SD: 0.0033
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

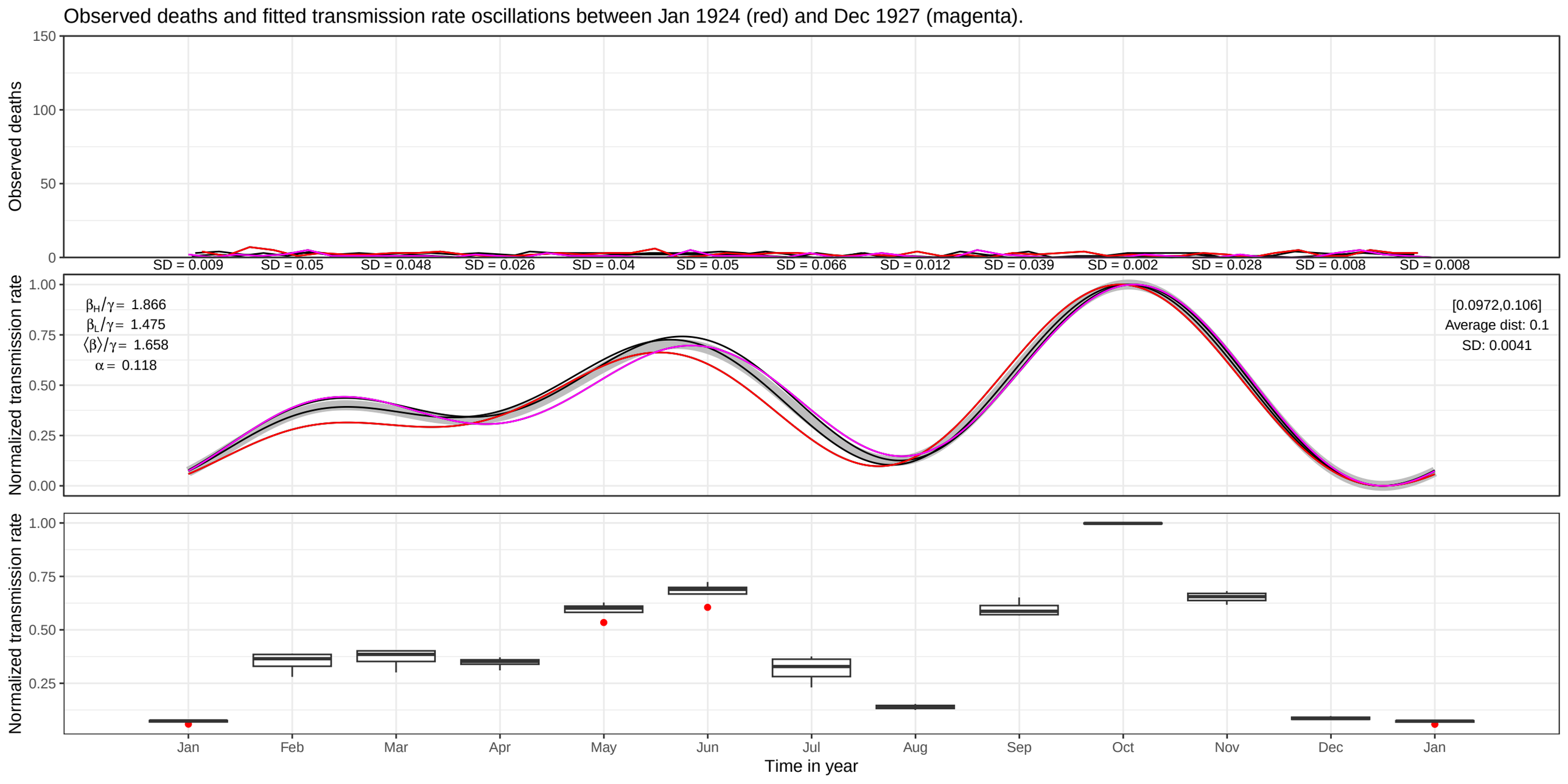

Observed deaths and fitted transmission rate oscillations between Jan 1896 (red) and Dec 1898 (magenta).
Observed deaths
150
100
50
0
SD = 0.007
SD = 0.009
SD = 0.024
SD = 0.061
SD = 0.094
SD = 0.061
SD = 0.192
SD = 0.152
SD = 0.01
SD = 0.021
SD = 0.049
SD = 0.056
SD = 0.007
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 2.289
βL/γ = 1.875
⟨β⟩/γ = 2.127
α = 0.097
[0.0567,0.0723]
Average dist: 0.065
SD: 0.0078
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

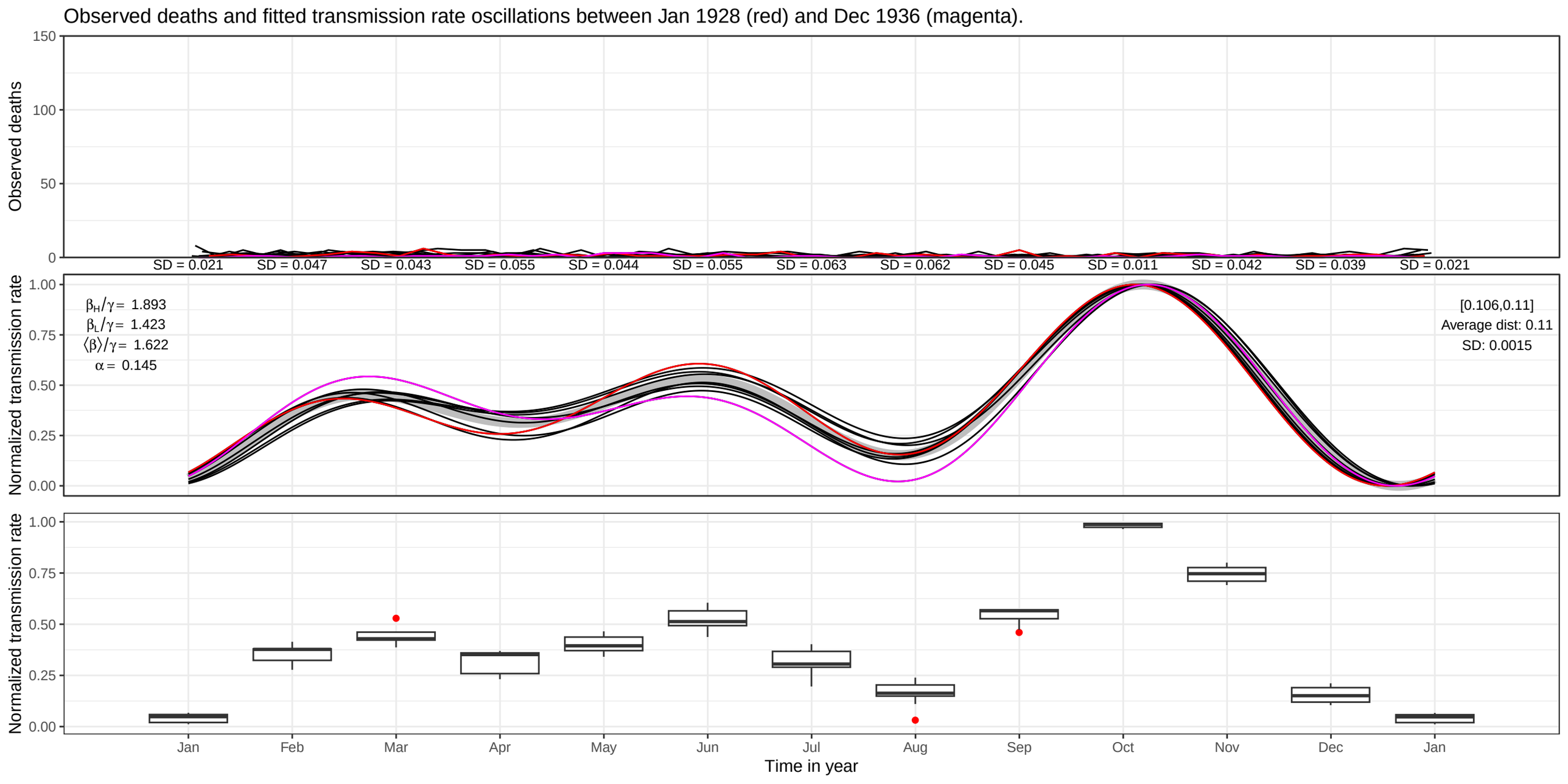

Observed deaths and fitted transmission rate oscillations between Jan 1899 (red) and Dec 1903 (magenta).
Observed deaths
150
100
50
0
SD = 0.066
SD = 0.158
SD = 0.085
SD = 0.052
SD = 0.05
SD = 0.012
SD = 0.038
SD = 0.062
SD = 0.101
SD = 0.094
SD = 0.144
SD = 0.112
SD = 0.066
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 2.296
$\beta_L/\gamma =$ 1.786
$\langle\beta\rangle/\gamma =$ 2.01
$\alpha =$ 0.127
[0.0689,0.0958]
Average dist: 0.084
SD: 0.012
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

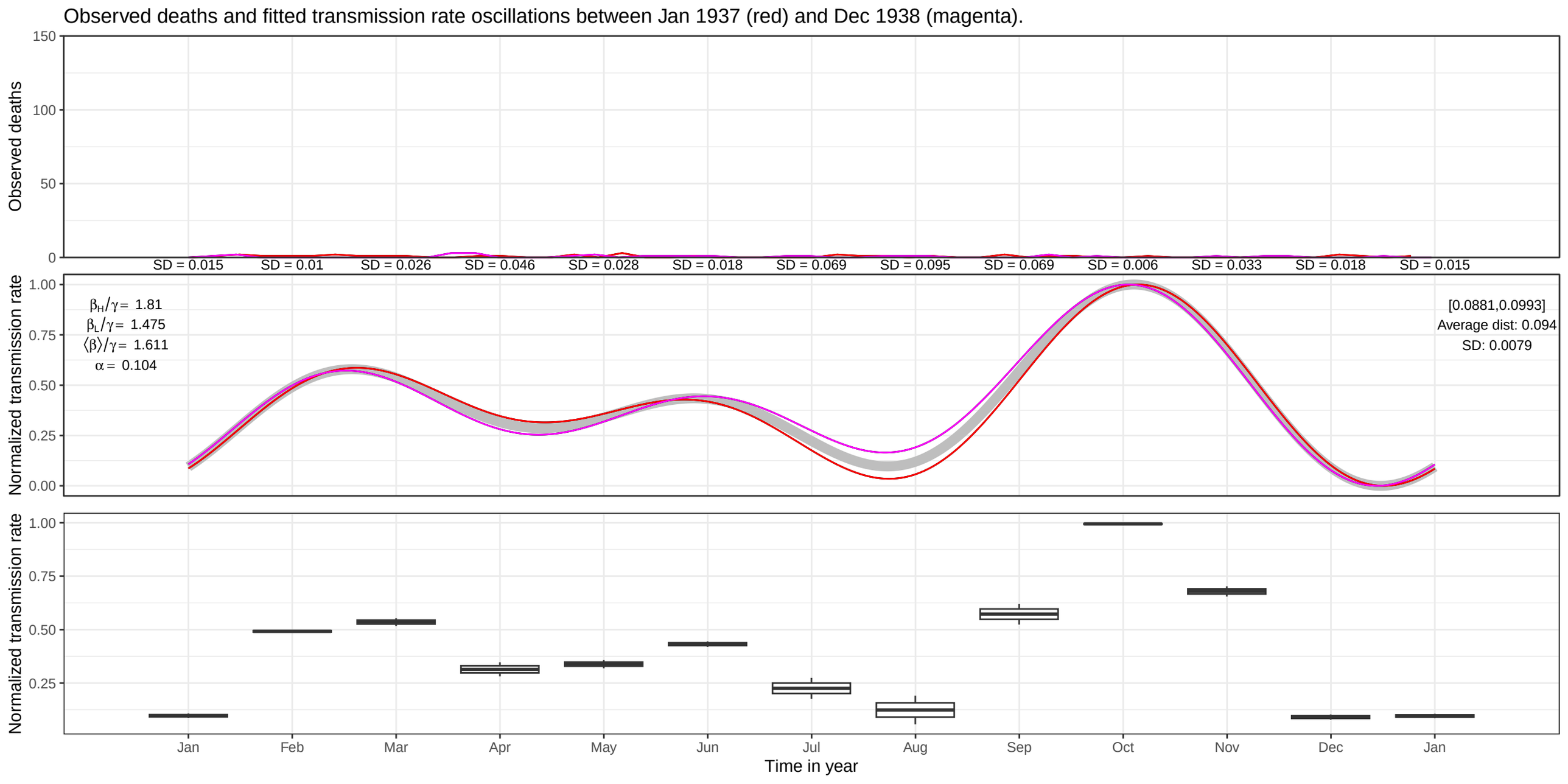

Observed deaths and fitted transmission rate oscillations between Jan 1904 (red) and Dec 1913 (magenta).
Observed deaths
150
100
50
0
SD = 0.038
SD = 0.061
SD = 0.066
SD = 0.077
SD = 0.055
SD = 0.046
SD = 0.033
SD = 0.049
SD = 0.062
SD = 0.012
SD = 0.071
SD = 0.035
SD = 0.038
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 2.124
$\beta_L/\gamma =$ 1.561
$\langle\beta\rangle/\gamma =$ 1.849
$\alpha =$ 0.152
[0.0781,0.108]
Average dist: 0.09
SD: 0.0092
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1914 (red) and Dec 1918 (magenta).
Observed deaths
150
100
50
0
SD = 0.016
SD = 0.053
SD = 0.051
SD = 0.035
SD = 0.06
SD = 0.058
SD = 0.043
SD = 0.038
SD = 0.022
SD = 0.001
SD = 0.007
SD = 0.01
SD = 0.016
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 2.023
βL/γ = 1.501
⟨β⟩/γ = 1.72
α = 0.152
[0.111,0.117]
Average dist: 0.11
SD: 0.0022
Normalized transmission rate
1.00
0.75
0.50
0.25
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1919 (red) and Dec 1923 (magenta).
Observed deaths
150
100
50
0
SD = 0.009
SD = 0.035
SD = 0.038
SD = 0.048
SD = 0.079
SD = 0.017
SD = 0.101
SD = 0.081
SD = 0.018
SD = 0.001
SD = 0.034
SD = 0.021
SD = 0.009
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 2.062
$\beta_L/\gamma =$ 1.477
$\langle\beta\rangle/\gamma =$ 1.685
$\alpha =$ 0.174
[0.107,0.121]
Average dist: 0.12
SD: 0.0054
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1924 (red) and Dec 1927 (magenta).
Observed deaths
150
100
50
0
SD = 0.009
SD = 0.05
SD = 0.048
SD = 0.026
SD = 0.04
SD = 0.05
SD = 0.066
SD = 0.012
SD = 0.039
SD = 0.002
SD = 0.028
SD = 0.008
SD = 0.008
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 1.866
βL/γ = 1.475
⟨β⟩/γ = 1.658
α = 0.118
[0.0972,0.106]
Average dist: 0.1
SD: 0.0041
Normalized transmission rate
1.00
0.75
0.50
0.25
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1928 (red) and Dec 1936 (magenta).
Observed deaths
150
100
50
0
SD = 0.021
SD = 0.047
SD = 0.043
SD = 0.055
SD = 0.044
SD = 0.055
SD = 0.063
SD = 0.062
SD = 0.045
SD = 0.011
SD = 0.042
SD = 0.039
SD = 0.021
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 1.893
βL/γ = 1.423
⟨β⟩/γ = 1.622
α = 0.145
[0.106,0.11]
Average dist: 0.11
SD: 0.0015
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1937 (red) and Dec 1938 (magenta).
Observed deaths
150
100
50
0
SD = 0.015
SD = 0.01
SD = 0.026
SD = 0.046
SD = 0.028
SD = 0.018
SD = 0.069
SD = 0.095
SD = 0.069
SD = 0.006
SD = 0.033
SD = 0.018
SD = 0.015
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 1.81
βL/γ = 1.475
⟨β⟩/γ = 1.611
α = 0.104
[0.0881,0.0993]
Average dist: 0.094
SD: 0.0079
Normalized transmission rate
1.00
0.75
0.50
0.25
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year